

SAHIL GOYAL

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 Scholar

 GitHub

 LinkedIn

 Medium

EDUCATION

Indian Institute of Technology Roorkee

2019 - 2024

B.S.+M.S. in Applied Mathematics

Department Rank 1 (Gold Medal)

Overall GPA: **9.44**/10

TECHNICAL SKILLS

Languages and Core Frameworks: Python, C++, JAX (Flax, Optax, Orbx, Chex), PyTorch

PUBLICATIONS

- ELT: Elastic Looped Transformers for Visual Generation.** [\[Paper\]](#)
Sahil Goyal*, Swayam Agrawal*, Gautham Anil, Prateek Jain, Sujoy Paul, Aditya Kusupati [\[ECCV '26\]](#)
- MaGNeTS: Masked Generative Nested Transformers with Decode Time Scaling.** [\[Paper\]](#) [\[Poster\]](#)
Sahil Goyal*, Debapriya Tula*, Gagan Jain, Pradeep Shenoy, Prateek Jain, Sujoy Paul [\[ICML '25\]](#)
- Design-o-meter: Towards Evaluating and Refining Graphic Designs.** [\[Paper\]](#) [\[Poster\]](#) [\[Survey\]](#)
Sahil Goyal*, Abhinav Mahajan*, Swasti Mishra, Prateeksha Udhayanan, Tripti Shukla, K J Joseph, Balaji Vasan Srinivasan. [\[WACV '25\]](#)
- Emotionally Enhanced Talking Face Generation.** [\[Paper\]](#) [\[Poster\]](#) [\[Code\]](#)
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [\[ICCV-W '23\]](#) [\[ACM MM-W '23\]](#)
- Emotional talking faces: Making videos more expressive and realistic.** [\[Paper\]](#) [\[Demo\]](#)
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Dhroov Goel, Sakshat Mali, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [\[ACM MM Asia '22\]](#) [\[Best Demo paper\]](#)

EXPERIENCE

- Pre-Doctoral Researcher at Google DeepMind** June 2024 - Present
- Research in **Efficient ML** (training-time compute efficiency, inference-time compute efficiency, and parameter efficiency) including contributions to **Veo** and **Gemini**, and publications at ICML [\[2\]](#) and ECCV [\[1\]](#).
- Research Intern at Adobe Research** May 2023 - July 2023
- Unified framework [\[3\]](#) for automated design evaluation and refinement, enabling iterative improvements for creators.
- Research Intern at MIDAS Lab @IITD** Sep 2021 - Jan 2023
- Multimodal talking face generation network [\[4, 5\]](#), winning **Best Demo Paper** award at ACM MM Asia.
- Machine Learning Intern at PANDO** March 2021 - April 2021
- Developed a **20x** faster 3-D space optimization algorithm for transport logistics.

KEY RESEARCH PROJECTS

- Efficient Video Tokenization** May 2026 - Present
Advisors: [Dr. Sujoy Paul](#), [Dr. Prateek Jain](#)
- Router based token dropping for **Gemini**.
- Elastic Looped Transformers for Visual Generation** Oct 2025 - April 2026
Advisors: [Dr. Aditya Kusupati](#), [Dr. Sujoy Paul](#), [Dr. Prateek Jain](#)
- Improved discrete and continuous diffusion models with **4x** less parameters (FID of **2.0** using just **~100M** params).
 - Developed novel **Intra-Loop Self-Distillation**, enabling **Elastic Inference** and dynamic compute scaling.
 - Achieved **3.5x** inference throughput and **~35%** lower peak HBM compared to standard transformer baselines.

Compression of Pre-trained LLMs

May 2025 - Present

Advisors: *Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Pradeep Shenoy*

- Optimizing **model compression** pipeline for **Gemini**, aiming for **minimal extra training cost**.
- Building and refining effective **structured-pruning** and **layer-wise distillation** techniques.
- Results show significant quality gains over iso-params and iso-FLOPs baselines.

Adaptive Compute for Visual Generation

June 2024 - Feb 2025

Advisors: *Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Aditya Kusupati*

- Developed an **adaptive computation** method [2] for multi-step generative models using **nested transformers**.
- Achieved **~3x** reduction in inference FLOPs for image/video generation.
- Implemented similar efficient techniques for Google's SOTA video generation model (**Veo**).

Automated Evaluation and Refinement of Graphic Designs

May 2023 - July 2023

Advisors: *Dr. Joseph KJ, Dr. Balaji Vasan Srinivasan*

- Built compact **Siamese networks** (**0.5M** params) to score **Graphic Designs**.
- Significantly outperformed GPT-4o and LLaVA-NeXT (7B) in ranking graphic designs.
- Tailored advanced **Genetic Algorithms** for SOTA **Layout refinement** (mean-IOU of **0.54** on Crello dataset).

Emotionally Enhanced Talking Face Generation

Sep 2021 - Jan 2023

Advisors: *Dr. Rajiv Ratn Shah*

- Developed the first **talking face generation** network [4, 5] that incorporated emotions.
- SOTA results on **CREMA-D** dataset (**83.2%** emotion classification accuracy and **FID** metric: **5.29**).
- Deployed the entire system as an interactive web interface.

SELECTED ACHIEVEMENTS

- PhD admits from UT Austin, EPFL, MILA, ELLIS. 2026
- **2 Patents** filed at US Patent Office on Efficient Visual Generation [**Patent 1**]. 2025
- Among **15**, from 18,000+ applicants, selected for Google's Pre-Doctoral Researcher Program. 2024
- **Department Gold Medal** (IIT Roorkee - Applied Mathematics). 2024
- **Best Demo Paper** at ACM MM ASIA conference. 2022
- **Gold Medal** at Inter IIT Tech Meet. 2022
- **Open Source: PyTorch-Vision** (debugged dataloaders, ported tests and updated docs). 2022
- Awarded ACM SigMM student travel grant for ACM MM Asia. 2022
- JEE Advanced Rank of **1583** among 1.5M students. 2019
- **600+** stars on  **GitHub**.

POSITIONS OF RESPONSIBILITY

- **Reviewer** at NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, AAMAS, WACV conferences. 2024 - Present
- **Mentor** at **Data Science Group IITR** 2020 - 2024
- **Undergraduate Teaching Assistant** at Academic Reinforcement Program IITR 2021

PROJECTS B.C. (BEFORE CHATGPT)

Model Extraction Attacks for Video Classification [**Code**]

Feb 2022 - March 2022

- Successfully extracted models across grey-box (5% original data) and black-box (synthetic data only) settings.
- Developed novel data augmentation, synthetic data generation and knowledge distillation techniques.
- Our method won **gold medal** at Inter IIT Tech Meet.

Global Wheat Detection Challenge 2021 [**Code**]

June 2021 - July 2021

- Implemented and benchmarked SOTA **object detection** models with diverse backbones for wheat-head detection.
- Improved accuracy to **56.2%** using training tweaks (data augmentation, pseudo labeling) and post-processing techniques (weighted boxes fusion, non-maximum suppression).